Manganese-rich food

Laboratory sample of pure manganese

Uses

Jefferson nickel

This US coin *from World War II was made using manganese and silver because nickel was in short supply.*

Railway tracks

These steel tracks *have manganese added to them to make them stronger.*

Dry cell battery

This battery *contains manganese dioxide.*

Unleaded petrol

This petrol *contains a manganese compound, which is less toxic than lead.*

Lascaux cave paintings, France

The black colour *comes from manganese dioxide.*

Purple glass bottle

This glass is coloured *by adding a manganese compound called permanganate.*

JOHAN GOTTLIEB GAHN

In 1774, Swedish chemist Johan Gottlieb Gahn discovered manganese by reacting manganese dioxide with charcoal – which contains carbon – under a lot of heat. The carbon took the oxygen away from the compound, leaving behind pure manganese.

from mussels, nuts, oats, and pineapples. The applications of manganese include its use in strengthening steel, which is used in making **railway tracks** and tank armour. Certain **dry cell batteries** carry a mixture containing manganese oxide. Manganese compounds are also added to **petrol** and used to clean impurities from **glass** to make it clear or to give it a purple colour. In prehistoric times, the compound manganese dioxide was crushed to make the dark colours used in **cave paintings**.